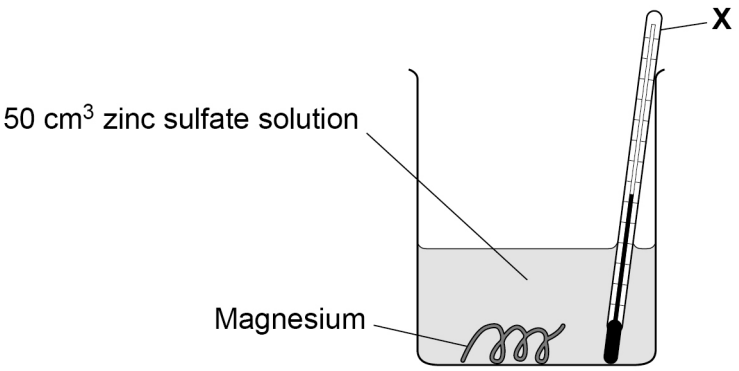
Do not write outside the box

0 2 A student investigated the temperature change when magnesium was added to zinc sulfate solution.

Figure 5 shows the apparatus.

Figure 5



0 2 . 1 Which piece of equipment is labelled **X**?

[1 mark]

Tick (✓) **one** box.

Beaker ☐

Ruler ☐

Thermometer ☐

0 2 . 2 Which piece of equipment is the best to use to measure volumes of solution?

[1 mark]

Tick (✓) **one** box.

Conical flask ☐

Evaporating basin ☐

Measuring cylinder ☐

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The student added 1.0 g of magnesium to 50 cm^3 of zinc sulfate solution and measured the temperature increase.

The student repeated the experiment two more times.

Table 1 shows the results.

Table 1

Temperature increase in °C			
Experiment 1	**Experiment 2**	**Experiment 3**	**Mean**
7.6	7.3	7.6	**Y**

0 2 . 3 Calculate value **Y** in **Table 1**.

[2 marks]

_____

_____

_____

Y = _____ °C

0 2 . 4 The student then added 1.2 g of magnesium to 50 cm^3 of zinc sulfate solution.

The temperature increased by 9.0 °C.

Calculate the temperature increase when the student adds 0.40 g of magnesium to 50 cm^3 of zinc sulfate solution.

[2 marks]

_____

_____

_____

Temperature increase = _____ °C

Do not write outside the box

0 2 . 5 What is the name given to a reaction which causes the temperature to increase?

[1 mark]

Tick (✓) **one** box.

Endothermic ☐

Exothermic ☐

Thermal decomposition ☐

0 2 . 6 The student repeated the experiment with 1.2 g of copper and 50 cm^3 of zinc sulfate solution.

The temperature did **not** increase.

Give **one** reason why.

[1 mark]

_____

_____

8